

Theory note S2 — The graph Laplacian, algebraic connectivity

$$\lambda_2,$$

and what the verification checks prove

Chang Young Ham

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Purpose. S2 summarises a whole network by one number, λ_2 . This note explains why that number is the right one: it proves the two facts the simulator relies on ($\lambda_2 > 0$ iff connected; $\lambda_2 \leq \frac{N}{N-1}\delta_{\min}$), states what the 36,090-graph consistency check verifies, and interprets the U-shaped λ_2 - ρ curve. Mathematics used: symmetric matrices, eigenvalues, quadratic forms, the Courant–Fischer characterisation.

1 From links to a matrix

Take N nodes. From S1, each pair (i, j) has a bit error rate $P_b(d_{ij}, J/S, \rho)$; the link is *up* if $P_b \leq \text{BER}_{\text{out}}$. Collect the result in the adjacency matrix

$$A_{ij} = \begin{cases} 1 & \text{link } (i, j) \text{ up, } i \neq j \\ 0 & \text{otherwise,} \end{cases}$$

symmetric with zero diagonal. The degree $d_i = \sum_j A_{ij}$ counts a node’s live links; $D = \text{diag}(d_1, \dots, d_N)$. The *graph Laplacian* is

$$L = D - A.$$

Its rows sum to zero by construction: $L\mathbf{1} = D\mathbf{1} - A\mathbf{1} = (d_i) - (d_i) = 0$.

2 Three facts about L

(a) **L is symmetric, so its eigenvalues are real.** D and A are symmetric. Hence L has N real eigenvalues, which we order $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$, and an orthonormal eigenbasis. `np.linalg.eigvalsh` exploits exactly this (the “h” is for Hermitian).

(b) **The quadratic form is a sum of squared differences over edges.** For any vector $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^\top L \mathbf{x} = \sum_i d_i x_i^2 - \sum_{i,j} A_{ij} x_i x_j = \sum_{\{i,j\} \in E} (x_i - x_j)^2 \geq 0. \quad (1)$$

Stop here and check: expand $\sum_{\{i,j\} \in E} (x_i - x_j)^2$, note that each edge contributes $x_i^2 + x_j^2$ (which sums to $\sum_i d_i x_i^2$ over all edges touching i) and $-2x_i x_j$ (which sums to $-\sum_{i,j} A_{ij} x_i x_j$ since each unordered edge appears twice in the double sum).

(c) **$\lambda_1 = 0$ always.** From (b), L is positive semidefinite, so all $\lambda_k \geq 0$; and $L\mathbf{1} = 0$ shows 0 is attained with eigenvector $\mathbf{1}$. Thus $\lambda_1 = 0$ carries no information about the graph — which is why the second eigenvalue is the one that matters.

3 Theorem: $\lambda_2 > 0$ if and only if the graph is connected

Theorem 1. *The multiplicity of the eigenvalue 0 of L equals the number of connected components of the graph. In particular $\lambda_2 > 0 \iff$ the graph is connected.*

Proof. Suppose $L\mathbf{x} = 0$. Then $\mathbf{x}^\top L\mathbf{x} = 0$, and by (1) every edge has $x_i = x_j$. Following edges, $\mathbf{x}$ is constant on each connected component. Conversely, any vector constant on components is in the null space (each row of L only mixes a node with its neighbours, which share its value). So the null space is spanned by the indicator vectors of the components, and its dimension is the number of components c . Hence $\lambda_1 = \dots = \lambda_c = 0$ and $\lambda_{c+1} > 0$; with $c = 1$ this reads $\lambda_2 > 0$. $\square$

This theorem is why the script can replace an explicit search over the graph by one eigenvalue. VERIFICATION 2 is a test of the implementation, not of the theorem: for every one of the 36,090 graphs generated during the sweeps it computes both $[\lambda_2 > 10^{-9}]$ and a breadth-first search, and demands they agree. Zero mismatches means the adjacency construction, the Laplacian, and the eigen-solver are consistent with a completely independent algorithm. (The 10^{-9} tolerance absorbs floating-point round-off; λ_2 of a disconnected graph comes back as $\pm 10^{-15}$, not exactly 0.)

4 Two exact benchmarks

Proposition 1 (Complete graph). *For K_N (every pair linked), $\lambda_2 = N$.*

Proof. $A = \mathbf{1}\mathbf{1}^\top - I$ and $D = (N-1)I$, so $L = NI - \mathbf{1}\mathbf{1}^\top$. On the subspace $\mathbf{x} \perp \mathbf{1}$ we have $\mathbf{1}\mathbf{1}^\top \mathbf{x} = 0$, so $L\mathbf{x} = N\mathbf{x}$: the eigenvalue N has multiplicity $N-1$. Together with $\lambda_1 = 0$ this is the whole spectrum, and $\lambda_2 = N$. $\square$

VERIFICATION 1 checks this for $N = 10, 20, 30$. It is the first thing to run when the eigen-solver or the sign convention of L is in doubt.

Proposition 2 (Fiedler’s degree bound). *Let $\delta_{\min} = \min_i d_i$. Then $\lambda_2 \leq \frac{N}{N-1} \delta_{\min}$.*

Proof. By the Courant–Fischer characterisation, $\lambda_2 = \min_{\mathbf{x} \perp \mathbf{1}, \mathbf{x} \neq 0} \mathbf{x}^\top L\mathbf{x} / \mathbf{x}^\top \mathbf{x}$, so any particular $\mathbf{x} \perp \mathbf{1}$ gives an upper bound. Let v be a node of minimum degree and take $\mathbf{x} = e_v - \frac{1}{N}\mathbf{1}$ (orthogonal to $\mathbf{1}$ since its entries sum to $1 - 1 = 0$). Every edge at v contributes $(x_v - x_j)^2 = 1$ and every other edge contributes 0, so $\mathbf{x}^\top L\mathbf{x} = d_v = \delta_{\min}$. Also $\mathbf{x}^\top \mathbf{x} = (1 - \frac{1}{N})^2 + (N-1)\frac{1}{N^2} = \frac{N-1}{N}$. Hence $\lambda_2 \leq \delta_{\min} \cdot \frac{N}{N-1}$. $\square$

A correction recorded honestly. The first draft of the script tested the *wrong* bound $\lambda_2 \leq \delta_{\min}$ and failed on 16,195 graphs. The failure was correct: for K_N , $\lambda_2 = N$ exceeds $\delta_{\min} = N-1$. The bound above is the right one and passes on all graphs. A verification that can fail is worth more than one that cannot.

5 Why λ_2 and not “number of live links”

Two graphs with the same number of links can differ completely: 20 links scattered evenly leave a network in one piece; 20 links concentrated on the only path between two clusters, if removed, split it. λ_2 sees the structure. It also has a dynamical meaning: for the consensus dynamics $\dot{\mathbf{x}} = -L\mathbf{x}$ (each node moves toward the average of its neighbours — the basic model of formation keeping), the disagreement decays as $e^{-\lambda_2 t}$. Small λ_2 therefore means *slow* coordination before it means *no* coordination, which is the early warning that a binary connected/disconnected flag cannot give.

6 Interpreting the sweeps

Collapse threshold (figure s2_lambda2_vs_js). For each jammer power the script lets the jammer choose the ρ that minimises mean λ_2 , then records the first J/S at which fewer than half of the 30 placements remain connected. The thresholds $-10/-8/-6$ dB for $N = 10/20/30$ say that, in this geometry, ten more nodes buy about 2 dB. The absolute values depend on the assumed path loss and reference distance; the spacing between curves is the robust statement.

The U-shape in ρ (figure s2_lambda2_vs_rho). S1 found the single-link worst case at $\rho^* \approx 0.14$; the network's worst case is an order of magnitude smaller, $\rho \approx 0.01$. The reason is the outage rule. A link is either up or down; once it is down, raising its BER further gains the jammer nothing. From the mixture formula, a link whose jammed hops are hopeless ($Q \rightarrow \frac{1}{2}$) has $P_b \approx \rho/2$, so it is down whenever $\rho > 2 \text{BER}_{\text{out}} = 2 \times 10^{-3}$. The jammer therefore does best with the *smallest* ρ that still trips the criterion — it makes a few hops hopeless on as many links as possible — which is why λ_2 falls as ρ decreases from 1. Below $2 \text{BER}_{\text{out}}$ the jammed hops are too rare to trip the rule and every link recovers, producing the rise on the left. The minimum sits just above 2×10^{-3} , as observed.

This is a property of the hard-threshold outage model, not of hopping itself. Replacing the threshold by a coded-packet error model would change ρ^* — one of the listed extensions.

7 Correspondence with the code

Mathematics	Code
$A_{ij} = [P_b \leq \text{BER}_{\text{out}}]$	<code>a = (ber <= BER_OUTAGE).astype(int)</code>
$L = D - A$	<code>lap = np.diag(a.sum(axis=1)) - a</code>
λ_2 (2nd smallest, symmetric)	<code>np.linalg.eigvalsh(lap)[1]</code>
connected? (independent)	<code>is_connected_bfs(a)</code>
$K_N: \lambda_2 = N$	<code>VERIFICATION 1</code>
$\lambda_2 > 0 \iff \text{connected}$	<code>VERIFICATION 2 (0 mismatches / 36,090)</code>
$\lambda_2 \leq \frac{N}{N-1} \delta_{\text{min}}$	<code>VERIFICATION 3</code>

8 Self-check

1. Show that $\mathbf{x}^T L \mathbf{x} = \sum_{\text{edges}} (x_i - x_j)^2$ for a triangle graph by writing out L explicitly.
2. Prove that $\lambda_1 = 0$ for every graph. Why is this uninformative?
3. Give the argument that the null space of L has dimension equal to the number of components.
4. Compute λ_2 of K_4 two ways: from the proposition and by writing L and finding its eigenvalues by hand.
5. Explain why the network's worst ρ is about $2 \text{BER}_{\text{out}}$ and not the S1 value 0.14. Which modelling choice is responsible?